

Crude oil price forecasting with multivariate selection, machine learning, and a nonlinear combination strategy

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ABSTRACT

Crude oil price forecasting has been one of the research hotspots in the field of energy economics, which plays a crucial role in energy supply and economic development. However, numerous influencing factors bring serious challenges to crude oil price forecasting, and existing research has room for further improvement in terms of an integrated research roadmap that combines impact factor analysis with predictive modelling. This study aims to examine the impact of financial market factors on the crude oil market and to propose a nonlinear combined forecasting framework based on common variables. Four types of daily exogenous financial market variables are introduced: commodity prices, exchange rates, stock market indices, and macroeconomic indicators for ten indicators. First, various variable selection methods generate different variable subsets, providing more diversity and reliability. Next, common variables in the subset of variables are selected as key features for subsequent models. Then, four models predict crude oil prices using common features as inputs and obtain the prediction results for each model. Finally, the nonlinear mechanism of the deep learning technology is introduced to combine above single prediction results. Experimental results reveal that commodity and foreign exchange factors in financial markets are critical determinants of crude oil market volatility over the long term, as observed in experiments conducted on the West Texas Intermediate and Brent oil price datasets. The proposed model demonstrates strong performance regarding average absolute percentage error, recorded at 2.9962% and 2.4314%, respectively, indicating high forecasting accuracy and robustness. This forecasting framework offers an effective methodology for predicting crude oil prices and enhances understanding the crude oil market.

List of abbreviations

ARIMA	Autoregressive integrated moving average	LGBR	Light Gradient Boosting Regressor
ARIMA	Autoregressive integrated moving average	LR	Logistic Regression
ARIMAX	Autoregressive integrated moving average with extra input	LSTM	Long short-term memory neural network
BiLSTM	Bi-directional Long Short-Term Memory	MAE	Mean absolute error
BPNN	Back Propagation Neural Network	MAPE	Mean absolute percentage error
CNN	Convolutional neural network	MARS	Multivariate adaptive regression splines
DM	Diebold–Mariano	MIC	Maximal information coefficient

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ELM	Extreme Learning Machine	MLP	Multilayer perceptron neural network
GARCH	Generalized Autoregressive Conditional Heteroskedasticity	RF	Random forest
GARF	Genetic algorithm based on random forest	RMSE	Root mean square error
GC	Granular computing	SVR	Support vector regression
GRU	Gated recurrent units	TCN	Temporal convolutional network
HGBR	Histogram-based Gradient Boosting Regression Tree	TLAC	Transfer Learning Based Analog Complexing Model
ISSA	Improved sparrow search algorithm	VR	Voting Regressor

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LASSO	Least absolute shrinkage and selection operator	WFTS	Weighted fuzzy time series
List of symbols.			
n	the number of data points	g_{t-1}	the hidden layer at the previous time
$\Theta(G)^p$	an order p function of this lag operator G	a_t, g_t, l_t	the input vector, hidden layer and update gate
Δ^d	the integration operator	d_t	the reset gate
$\Phi(G)^q$	the polynomial function for the error	$\sigma(t)$	the sigmoid function
b_t	the predicted value of time step target	$\tanh(t)$	the hyperbolic tangent function
$\sum_{j=1}^m \beta_j a_t^j$	the exogenous variables and coefficients	s	the bias vector
m_0	the whole number of exogenous attributes	$g_t, \hat{g}_t$	the output and candidate vector of GRU
m_j, c_j	the input and bias of j -th neuron	a_t, a_{t-1}	the present cell state and the previous cell state
p_{ij}	the weight between the i -th and j -th input neuron	h_t, h_{t-1}	the current hidden state and the previous hidden state
$q_j, f(s_j)$	the j -th neuron's outputs and activation function	r_t	the input vector
w_k	the k th output of the neurons	g_t	the forget gate output
n_0	the number of neurons in the hidden layers	p_t	the input gate output
p_{jk}	the weight between neurons in the j -th hidden layer and the k th output layer	q_t	the output gate output
w_k	the k th output of the neurons	I	the size of sample
a_i^{m-1}	the i -th input feature map	$P_i, \hat{P}_i$	the actual and forecasted values
s_i^m, p_{ji}^m	the convolution kernel biases and weights	P	the forecasting error function
w_i^m	the i -th output map	r_t^1, r_t^2	the predicted errors
$\otimes$	the convolution operation	D	the length of forecasting values
u_A, σ_A^2	the mean and the variance	$\bar{e}$	the mean of the loss function difference
a_j, w_j	the input and output features	$\widehat{f_e}(0)$	the zero spectral density
ε	the infinitesimal value	α	the significance level
γ, β	the scale factor and the movement factor	E_b, E_a	the MAPE value of the developed and compared model
n_1	the sample number for the current batch	I_{MAPE}	the extent of improvement

1. Introduction

1.1. Background

Crude oil is universally acknowledged as a crucial strategic asset by nations, and its pricing is often viewed as the critical benchmark for tracking commodity prices. The substantial fluctuations in crude oil prices significantly impact the global economy. Furthermore, extensive literature has long demonstrated that oil price fluctuations significantly affect external balances, economic growth, and price levels (Backus and Crucini, 2000; Aguiar-Conraria and Wen, 2007; Bachmeier and Cha, 2011; Kilian, 2009; Hamilton, 2008; Natal, 2012; Jo, 2014). In this context, accurate forecasting of crude oil prices is crucial for ensuring the stable operation of the crude oil trading market. A well-developed and precise forecasting model for crude oil prices enables market participants to manage risks more effectively and provides investors with scientifically grounded decision-making tools, holding significant research value (Morana, 2013; Wu and Zhang, 2014).

Crude oil price forecasting is widely recognized as a complex task due to the non-linearity and instability of crude oil prices and the influence of various factors. Policymakers, central banks worldwide, as well as investors and researchers, are keen on making accurate forecasts

regarding crude oil prices (Barsky and Kilian, 2004; Wan and Kao, 2015; Picciolo et al., 2017; Safari and Davallou, 2018; Zhao et al., 2018; Yang et al., 2024b). On the one hand, many studies have adopted classical statistical models and current mainstream artificial intelligence (AI) models as the main prediction tools and have mainly used linear combinations in the combined models, which makes these prediction tools focus more on capturing linear features but leave much to be desired in terms of their ability to capture non-linear features. On the other hand, with the increased financialization of the crude oil market, scholars have begun to study the impact of commodities and financial markets on oil prices (Akram, 2009; Tang and Xiong, 2012; Büyükhahin and Robe, 2014; Zhao et al., 2022). Recent studies have shown that financial market data can provide important forecasting information. Since critical underlying data such as supply, demand, and inventories are usually only available monthly or annually, financial market data available in real-time has a distinct advantage. Given the importance of timely access to data for analyzing crude oil price volatility, this study will focus on the impact of financial market factors on crude oil prices. Therefore, this study concentrates on forecasting crude oil prices and examining the influence of financial market factors on these prices, proposing a framework for nonlinear combinatorial forecasting.

1.2. Literature review

When it comes to time series, linear and nonlinear models are needed to achieve a forecasting process employing historical data (Yang et al., 2021a). In general, there needs to be more than one single model to assemble all relationships within empirical time series. Moreover, no prediction model could always yield wonderful results under varying situations from the recent literature, as each problem in the real world is of some particular intricacies (Khashei and Bijari, 2011; Heng et al., 2022). Fortunately, a combined model carries the potential to be a valid medium for improving the ultimate prediction performance, as well as facilitating the capturing of different types of relationships. Thus, the compound of different methods is highly recommended rather than employing only a single model (Timmermann, 2006).

Research on forecasting has primarily focused on traditional econometric models and machine learning models. Traditional econometric models used for crude oil price forecasting include Autoregressive Integrated Moving Average (ARIMA) models (Mohammadi and Su, 2010), Generalized Autoregressive Conditional Heteroscedasticity (GARCH) family models (Ma et al., 2019), Machine learning models comprise Neural network (NN) models (Huang and Wang, 2018), Extreme Learning Machine (ELM) model (Weng et al., 2021), Long Short-Term Memory model (LSTM) (Xu et al., 2024) and others (Weng et al., 2022; Yang et al., 2023; Su et al., 2023). Both types of models have areas that can continue to be optimized. Traditional econometric models rely on predetermined assumptions and demonstrate weak performance in capturing nonlinear features. In contrast, machine learning models are prone to overfitting and issues related to parameter sensitivity (Yang et al., 2021b). Therefore, some scholars have gradually turned their attention to combined model in the hope of leveraging the potential of combinatorial models to improve the predictive accuracy and stability of the models.

Combined forecasting studies explore the integration of multiple competing forecasts to produce a composite forecast that is superior to individual forecasts. Over the past decades, various combination methods have been proposed, including simple averaging, trimmed mean, and median (Xiao et al., 2012; De Oliveira et al., 2017). As AI methods continue to evolve, there has been a growing trend towards applying optimization algorithms to determine the optimal weights of models. For instance, Wang et al. obtained the optimal weighting vectors by the artificial bee colony (ABC) technique (Wang et al., 2020). Cao et al. (2024) introduced a combined optimization mechanism centred around an improved sparrow search algorithm (ISSA) to harmonize the forecasting outcomes of three individual models (Cao et al., 2024).

Research indicates that linear combinations of empirical and deterministic forecasts can improve predictive accuracy; however, this enhancement is only realized when the individual forecasts are both independent and comparable (Wang et al., 2020). Table 1 shows the common methods and models used in existing studies.

As the factors influencing crude oil prices are deeply explored, univariate forecasting models are gradually replaced by multivariate forecasting models. Although multivariate models can provide more comprehensive analyses, the increase of variables also brings concerns about redundant information, computational efficiency, and so on. Therefore, numerous techniques for feature selection have been used in the existing prediction literature. Feature selection is capable of identifying and removing a significant number of irrelevant and redundant features. In most research related to forecasting crude oil prices, feature selection is commonly employed for data reduction, as it helps to maintain the original features, facilitating improved model interpretability (Yang et al., 2021b). For instance, for the electricity market, some authors introduced an adjusted edition of mutual information (MI) analysis to achieve variable selection (Abedinia et al., 2016). Sermpinis et al. (2018) employed least absolute shrinkage and selection operator (LASSO) variable selection techniques to model market implied ratings. Similarly, some authors also employed multivariate adaptive regression splines (MARS) to capture significant attributes for stock index forecasting (Dai et al., 2013). Some scholars also used similar feature selection methods to refine exogenous variables (Xiao et al., 2012; Guo et al., 2022). Table 2 shows some of the literature in the prediction category using the feature screening approach.

1.3. Main research gaps

Existing crude oil price forecasting frameworks commonly face competition among various feature screening methods. This competition

Table 2

Forecasting research with different feature selection method.

Authors	Forecasting target	Method	Categories
Xu et al. (2024)	Crude oil price volatility	LASSO	Embedded
Pan et al. (2024)	Ultra-short-term electric load	MIC	Filter
Guo et al. (2022)	Crude oil price	RFE	Wrapper
Desbordes et al. (2017)	Outcome prediction	GARF	Embedded
Ludwig et al. (2015)	European Power Exchange electricity spot prices	RF	Embedded
Dai et al. (2013)	Stock index	MARS	Model construction
Xiao et al. (2012)	Crude oil price	PCC	Filter

Note: Abbreviations appearing in this tables can be seen in the abbreviated list. And the specific meaning of Filter, Embedded, and Wrapper can be seen in subsection 2.1.

often leads researchers to select a subset of features from one screening method for input into subsequent models. However, this approach is susceptible to the limitations of relying on a single method. This study employs a collaborative approach in which each feature screening method evaluates a subset of features from diverse theoretical perspectives. The core subset of features is identified by examining common variables across these subsets. This methodology, characterized by the cooperation between different screening methods, is implemented in the present study.

Additionally, existing combinatorial models often assume that the contributions of models are linearly combined, which restricts their ability to capture complex data and non-linear relationships, potentially leading to information loss and reduced predictive power. In contrast, non-linear combinations are better suited to capturing complex interactions and non-linear features between models, thus enhancing

Table 1

Overview of forecasting models.

Authors	Forecasting target	Model	Model type	Advantages	Disadvantages
Mohammadi and Su (2010)	Crude oil price	ARIMA	Statistical model	Statistical models are simple to implement	They cannot deal with the non-linear features due to the pre-assumed linear form of these models
Ma et al. (2019)	Crude oil price volatility	GARCH			
Huang and Wang (2018)	Global crude oil price	BPNN	Machine learning model	They can recognize non-linear relationships in a dataset regardless of whether they are known.	The drawbacks of AI models such as local minimal point, overfitting problems, limited by insufficient data and requiring larger samples etc.
Weng et al. (2020)	Gold price	ELM			
Weng et al. (2021)	Crude oil price volatility	ELM			
Xu et al. (2024)	Crude oil price volatility	LSTM			
Weng et al. (2024)	Energy consumption	XGBoost			
Yang et al. (2024a)	Carbon price	XGBoost			
Yao et al. (2024)	Crude oil price	CNN-BiLSTM	Combined Model	By integrating the unique strengths of various models, they can offer greater prediction accuracy and stability compared to other models.	The researcher designs each hybrid model to tackle a particular challenge, achieving outstanding results for specific real-world scenarios. However, this approach might not deliver the same level of performance when used for different types of problems.
Xiao et al. (2012)	Crude oil price	TLAC			
De Oliveira et al. (2017)	Time series	ARMA			
Wang et al. (2020)	Crude oil price	LR, ANN, SVR			
Hao et al. (2023)	Exchange rates	ELM			
Lin et al. (2023)	Carbon emissions	LSTM, GRU			
Pei et al. (2024)	Wind speed	ARIMA, ELM, SVR, LGB, LSTM			
Bâra and Oprea (2024)	Bitcoin price	RF, XGB, LGBR, HGBR, VR			
Cao et al. (2024)	Electricity load	TCN, BiLSTM-attention, ISSA-WFTS			

Note: Abbreviations appearing in this tables can be seen in the abbreviated list.

predictive performance and flexibility while minimizing information loss.

1.4. Main contributions

Inspired by the previous discussion, this study proposes a common variable heterogeneous combination approach to improve forecasting accuracy. Firstly, four types of financial market indicators are considered. Subsequently, five feature selection techniques are used to construct feature subspaces independently, which helps to identify the key variables affecting crude oil prices. The final forecasts are then determined by aggregating the forecasts of each model within the covariate feature space and optimizing the weights using the LSTM model. The aim of this study is to explore the optimal forecasting combinations to achieve more accurate forecasting performance. The contributions of this study are summarized below.

- (1) Investigating four types of financial market indicators on crude oil prices and explore their core influences;
- (2) Integration of multiple feature selection techniques to address bias stemming from the constraints of a single method.
- (3) Introducing a variety of traditional statistical and machine learning models to capture both linear and nonlinear features of crude oil prices. This work introduces ARIMAX, GRU, LSTM, and MLP models across different variable subsets to effectively leverage the linear and nonlinear characteristics of crude oil prices. By maximizing information extraction, these models contribute to the enhanced prediction capabilities of the combined model.
- (4) To overcome the limitations of linear combination methods, a common variable heterogeneous combination framework is proposed based on the nonlinear mechanism to combine single prediction results.

The subsequent sections outline the organization of this study. [Section 2](#) outlines the methodology employed. Moreover, [Section 3](#) presents the proposed LSTM-based combination mechanism. [Section 4](#) provides a comprehensive overview of the experiments and analysis of results. Subsequently, [Section 5](#) offers a discussion. Finally, [Section 6](#) contains the conclusions of the study.

2. Methodology

This chapter discusses the techniques employed in detail, involving subset constructing and feature selection approaches, and forecasting models for hybridization, such as multilayer perceptron (MLP) neural network. Then, a particular combined model to improve the forecasting performance is demonstrated.

2.1. Subset constructing and feature selection

Subset constructing (SC) is newly developed to imitate human thinking mode to address intricate issues. There are three fundamental notions involved, including subset, subset space, and hierarchies ([Deng et al., 2016](#); [Wang et al., 2020](#)). Each subset space assembles diverse but participant variable types as a hierarchical structure. Given that n is the magnitude of the feature subset chosen to tackle the prediction work, it is responsible for distributing the whole solution space to $n+1$ subset space, which contains n factors as well as the dependent variable.

On account of the association between feature selection approaches and different forecasting models, feature abstraction methods can be categorized into three types: covering filter, wrapper, and embedded methods. Filter methods solely take into account the relationship between features and the target variable, enabling the computation of critical metrics that are crucial for the application of filtering techniques ([Cai et al., 2018](#)). As a result, compared with wrapper methods, filter

mechanisms carry less computational cost. A wrapper is another way to consider the chosen attribute sets as a search task. Initially, it prepares a subset of some variables prior to comparison and evaluation. Besides, the embedded methods ([Efron et al., 2004](#)), the most familiar of which is the regularization method, aim to ascertain what feature contributes more to further accuracy.

2.1.1. Filter

(a) Maximal information coefficient (MIC)

As is well-known, mutual information (MI) acts as a method to estimate the codependence between two variables. Firstly, MIC devotes to find an optimum discretization method ([Reshef et al., 2011](#)). Then, it transforms the mutual information value into an evaluation approach whose interval is vary from 0 to 1. If MIC is nearer to 1, then there is a relatively firmer correlation between two attributes. Meanwhile, the two variables tend to be independent if MIC is closer to zero ([Pan et al., 2024](#)).

(b) Pearson correlation coefficient (PCC)

It is an approach widely employed to estimate the degree of linearity between two attributes, and whose value is between -1 and 1 . Especially, a vital mathematical property of Pearson correlation coefficient is that changes of the coefficient are not affected by the transformation of position and size of factors. The equation is expressed as:

$$r_{ab} = \frac{n \sum a_i b_i - \sum a_i \sum b_i}{\sqrt{n \sum a_i^2 - (\sum a_i)^2} \sqrt{n \sum b_i^2 - (\sum b_i)^2}} \quad (1)$$

Here, a_i , b_i and n indicate the a-variable, b-variable and the number of data points respectively.

2.1.2. Wrapper

(a) Recursive feature elimination (RFE)

This technology performs by recursively shortening the magnitude of attribute sets inspected. After being assigned a weight, each feature is examined to determine whether it carries the smallest absolute weight or not. The one who is the smallest will be removed from this feature set. Moreover, it is not until the number of residual variables achieves the supposed one that will quit reciprocal recursion.

2.1.3. Embedded

(a) Least absolute shrinkage and selection operator (LASSO)

As introducing a penalty function and impacting some coefficients, LASSO emerges as a more precise method. Most importantly, the model is good at astringing the subset. Meanwhile, this technology also does well in identifying the most important variables from a substantial amount of candidate features. Possibly, LASSO also helps in addressing the issue of multicollinearity ([Ludwig et al., 2015](#); [Sermpinis et al., 2018](#)).

(b) Random forest (RF)

RF is treated as a strong algorithm since it incorporates some random trees as well as assembling their prediction results through averaging ([Svetnik et al., 2003](#)). RF is good at calculating the significance of an individual factor. After ranking the significance of each variable, the top one is expected to be chosen. There are two approaches for feature abstraction, namely mean decrease impurity and mean decrease accuracy. In this paper, the mean decrease impurity is employed.

2.2. Auto-regressive integrated moving average (ARIMA)

ARIMAX model is a specific kind of ARIMA that takes account of exogenous variables. There are four main components contained, including the autoregressive, integrated, moving average, and exogenous variable parts. In this model, p is a parameter that stands for the time lag number used within prediction, as well as the autoregressive order. The number of error terms' time lags is denoted by q , which is used to determine how the past errors are treated during the current prediction. Then, the parameter d presents the order of integration difference that is employed to release non-stationary influence before forecasting. The exogenous variables indicate the attributes which affect the target while not influenced by other features. A classic ARIMAX (p, d, q) is expressed as follows:

$$\Theta(G)^p \Delta^d b_t = \Phi(G)^q \Delta^d \varepsilon_t + \sum_{j=1}^{m_0} \beta_j a_t^j \quad (2)$$

In this equation, $\Theta(G)^p$ acts as an order p function of this lag operator G , which corresponds to the autoregressive process. Δ^d is the integration operator, corresponding to the integrated part. $\Phi(G)^q$ denotes the polynomial function for the error, according to the moving average model. b_t is the predicted value of time step target. For the right part of formula, the $\sum_{j=1}^{m_0} \beta_j a_t^j$ is the exogenous variables and coefficients and m_0 is the whole number of exogenous attributes.

2.3. Multilayer perceptron neural network (MLP)

As a feedforward neural network, MLP, which uses parallel advanced structures, is widely introduced to forecast target attributes. During this period, it works by processing neurons through input, hidden, and output layers. For MLP, its output of hidden neurons is calculated by the following summation equation:

$$m_j = \sum_{i=1}^n p_{ij} x_i + c_j \quad (3)$$

$$q_j = f(m_j) = (1 + e^{-m_j})^{-1} \quad (4)$$

Here, in the hidden layer, m_j and c_j denote the input and bias of j -th neuron, respectively. p_{ij} indicates the weight between the i -th and j -th input neuron. Otherwise, q_j presents j -th neuron's outputs and $f(m_j)$ works as activation function. The following formula gives the outputs of the last layer:

$$W_k = \sum_{j=1}^{n_0} p_{jk} q_j + c_k \quad (5)$$

Where, w_k and n_0 are the k th output of the neurons and the number of neurons in the hidden layers, respectively. And p_{jk} denotes a weight between neurons in the j -th hidden layer and the k -th output layer. Though MLP find parameters by using the backpropagation algorithm, it is likely to be restricted to local optimums as well.

2.4. Convolutional neural network (CNN)

Equipped with multi-layer structures and multiple convolutional kernels, the CNN has the ability to extract deep-level and multi-angle of the initial signal. Meanwhile, CNN performs better on account of its superior discrimination and generalization capability compared with input features designed by hand. Generally, the composition of the CNN contains the input layer, convolution layer, pooling layer, as well as full connection and output layer. Moreover, the batch normalization layer is also included, aiming to facilitate the model's convergence and generalization ability.

Firstly, the output multiple feature maps are obtained by using a detailed number and size of convolution kernels. In this way, for input map, the convolution operation is presented as:

$$w_i^m = \sum_{j=1}^l p_{j,i}^m \otimes a_i^{m-1} + s_i^m \quad (6)$$

Here m indicates the number of layers and a_i^{m-1} is the i -th input feature map; s_i^m and $p_{j,i}^m$ are the convolution kernel biases and weights, respectively; w_i^m denotes the i -th output map; $\otimes$ is the convolution operation. Furthermore, in order to tackle more extended time series, the dilated operation is added to enlarge the model's reception domain.

With the aim to avoid the gradient disappearance of neural network, the batch normalization layer is introduced to guarantee that the activation input value exists at the area where the nonlinear function is impressive to the input. Consequently, a comparatively big gradient could be reached. The calculation method of batch normalization layer is described as follows:

$$u_A = \frac{1}{n_1} \sum_{j=1}^{n_1} a_j \quad (7)$$

$$\sigma_A^2 = \frac{1}{n_1} \sum_{j=1}^{n_1} (a_j - u_A)^2 \quad (8)$$

$$\hat{a}_j = \frac{a - u_A}{\sqrt{\sigma_A^2 + \varepsilon}} \quad (9)$$

$$w_j = \gamma \hat{a}_j + \beta \equiv AN_{\gamma, \beta}(a_j) \quad (10)$$

where u_A is the mean and σ_A^2 present the variance; a_j denotes the input features of batch normalization layer. Also, w_j is the output features of the batch normalization layer. ε denotes the infinitesimal value; γ is scale factor and β is the movement factor; n_1 denotes the sample number for the current batch. In the next period, the activation function is employed to achieve the nonlinear mapping for the input data and output data. Followed by a pooling layer, the activation layer is commonly used to reduce the input size, avoid overfitting as well as accelerate the computing speed. Finally, the output is obtained while the features completed with pooling and convolution are flattened into the fully connected layer.

2.5. Gated recurrent units (GRU)

GRU neural network is first developed to address problems associated with machine translation, which is also employed in finance later. Also, the GRU is allowed to tackle the issues of gradient disappearance as well as gradient explosion, which arise in RNNs. What's more, it simplifies the LSTM computation complex. The equations for controlling GRUs are shown below:

$$l_t = \sigma(M_p \cdot a_t + N_p \cdot g_{t-1} + s_p) \quad (11)$$

$$d_t = \sigma(M_t \cdot a_t + N_t \cdot g_{t-1} + s_d) \quad (12)$$

$$g_t^* = \tanh[M_g \cdot a_t + N_g(d_t^* g_{t-1})] \quad (13)$$

$$g_t = (1 - p_t) * g_{t-1} + l_t^* h_t^* \quad (14)$$

In equations mentioned above, upper-case variables indicate matrices and lower-case attributes represent vectors. At time t , a_t and g_t are the input vector and hidden layer, respectively. l_t is the update gate. g_{t-1} stands for the hidden layer at the previous time. d_t denotes the reset gate. The SoftMax functions $\sigma(t)$ and $\tanh(t)$ are the sigmoid function and hyperbolic tangent function, respectively. M and N are the weights for each layer, while s is the bias vector. g_t and g_t^* represent the output and candidate vector of GRU, respectively.

Most importantly, the GRU achieves the ability to perform remembering and forgetting steps simultaneously. In the formula for g_t , the p term that decide the message to be forgotten is related to the $(1-p)$ term

of the selected memory, while the memory and forgetting weights keep constant. It is this design that allows the GRU structure briefer than that of LSTM, while there is no reduction in accuracy.

2.6. Long short-term memory neural network (LSTM)

It is widely acknowledged that the main weakness of RNN is the exploding gradient and vanishing gradient. Fortunately, LSTM is brought up to deal with this problem. It contains a special input. Besides, to determine which information is to be reserved, the forget gate is proposed to possess gradients to flow more easily. In this way, it preserves wide ranges of output after settling common RNN difficulties. Compared with traditional neural networks, LSTM is undoubtedly good at remembering a much long sequence of data points. Moreover, they can also break through the restriction of vanishing gradient. In consequence, LSTM has been widely employed for intricate forecast issues, such as time series prediction. In Jeenanunta et al., 2018, the authors use the LSTM to forecast stock prices. Also, compared to ARIMA in Siami-Namini et al. (2018) for time series prediction, the LSTM demonstrated superior performance. In addition, with respect to speech recognition problems, the LSTM showed more superior property than RNN.

As for the structure of LSTM, there are several layers, containing input layer, hidden layer, and output layer, stacked up in the LSTM. When training the model, every layer is vested with weights and biases. Thanks to its property for backpropagation, the LSTM minimizes the error rate in the stage of minimizing the loss function. Consequently, the weights and biases are updated. LSTM has three gates: an input gate, an output gate, and a forget gate. Here at time t , a_t is present cell state, a_{t-1} is previous cell state, r_t is the input vector, h_t is the current hidden state, h_{t-1} is the previous hidden state, g_t is forget gate output, p_t presents input gate output and q_t denotes output gate output.

In LSTM, gate functions serve on adjusters which are applied to add or delete information. These gates aim to figure out which data is significant and remove unconsidered data. The input gate is regarded as the sigmoid layer. After receiving the message from the current input and previous hidden state, we now implement the sigmoid function. Using forgetting gates to determine which message could be remembered and removed. As a result of the forget gate, LSTM is not only capable of resetting its state but can also determine the duration of value in this cell. As shown in Eq. (15), forget gate take h_{t-1} and r_t as input and its output is vary from 0 to 1. If output value is near 0, they are likely to be irrelevant while considered as significant if it closes to 1. Later, the cell state is to sustain the information. Firstly, according to Eq. (16), the update value will be determined by input layer. After that, a new vector values a_t^* is added by tanh layer. The state is given by Eq. (17). Then, they are composed to update the state.

$$g_t = (w_g \cdot [h_{t-1}, r_t] + e_g) \quad (15)$$

$$p_t = (w_p \cdot [h_{t-1}, r_t] + e_p) \quad (16)$$

$$a_t^* = \tanh(w_a \cdot [h_{t-1}, r_t] + e_a) \quad (17)$$

In this way, using new cell state a_t , the old state a_{t-1} is updated. As depicted in Eq. (18), the old state a_{t-1} is multiplied by g_t to miss irrelevant information. The new values are scaled and then reset every cell state value.

$$a_t = g_t^* a_{t-1} + p_t^* a_t^* \quad (18)$$

The sigmoid layer determines the output and is employed tanh, by which its value varies from -1 to 1 . At the last step, with the help of the sigmoid gate, the output is gained, whose function is shown as Eqs. (19) and (20).

$$q_t = (w_q \cdot [h_{t-1}, x_t] + e_q) \quad (19)$$

$$h_t = q_t^* \tanh(a_t) \quad (20)$$

3. Construction of the combined model

Given that many existing combination methods utilize linear techniques, this section develops a hybrid model employing a nonlinear weighted combination approach. The model comprises three components: a feature selection module, a forecasting module, and a combination module. The framework is illustrated in Fig. 1, with further details provided below.

Module 1: Feature selection.

As previously discussed, utilizing full-dimensional attributes can increase computational complexity and decrease prediction accuracy. Additionally, relying on a single feature selection method introduces bias. Five feature selection techniques are employed, categorized into three types: filtering methods, wrapping methods, and embedding methods. Since these techniques produce different feature subsets, identifying common features and integrating the three evaluation criteria enables the creation of a more reliable feature set.

Module 2: Prediction of single model.

At this stage, the linear model ARIMAX and the nonlinear networks MLP, GRU, and LSTM are included in this study as the underlying methods of the hybrid model. Sufficient evidence suggests that hybrid models have the potential to be an effective means to improve the final prediction performance and help extract different types of relationships. At this stage, the common variables are input into the four models to obtain the corresponding prediction results, and the detailed process is shown in Fig. 1.

Module 3: Nonlinear combined technique.

After successfully completing individual method prediction employing common features, a non-linear combined technique based on LSTM is introduced to composite previous results. Practically, the substance of this approach is to realize the second forecasting for different types of messages. What's worth noting, the input in **Module 3** is equal to four new variables obtained by four individual methods within **Module 2**. In doing so, the first 70% of the forecasting results are applied to the training set, and the remaining 30% are considered as the test dataset.

4. Experiments and results analysis

This section includes the experimental process and analysis of the results, specifically covering data and feature selection, evaluation metrics, implementation of the three sets of experiments, summary of the experiments, and simulated trading.

4.1. Data and feature selection

4.1.1. Data description

The futures prices of West Texas Intermediate (WTI) and Brent crude oil are widely recognized as benchmarks for evaluating the global crude oil market due to their significant representativeness and broad influence. As crude oil is primarily traded in U.S. dollars, fluctuations in exchange rates impact its relative price and, consequently, the costs associated with exporting and importing crude oil. Additionally, stock market performance reflects investor confidence, considerably affecting crude oil prices. Macroeconomic monetary policy also plays a crucial role in influencing crude oil market movements. In addition, considering the impact of historical data on time series data, historical prices of WTI crude oil and Brent crude oil lagged by one day are included in this experiment. These two lagged one-day price data will be represented by P_{t-1}^{WTI} and P_{t-1}^{Brent} in the subsequent analyses.

This study incorporates natural gas, gold, and silver to provide a more comprehensive analytical perspective on the crude oil market. As a related energy commodity, natural gas can reflect changes in the crude

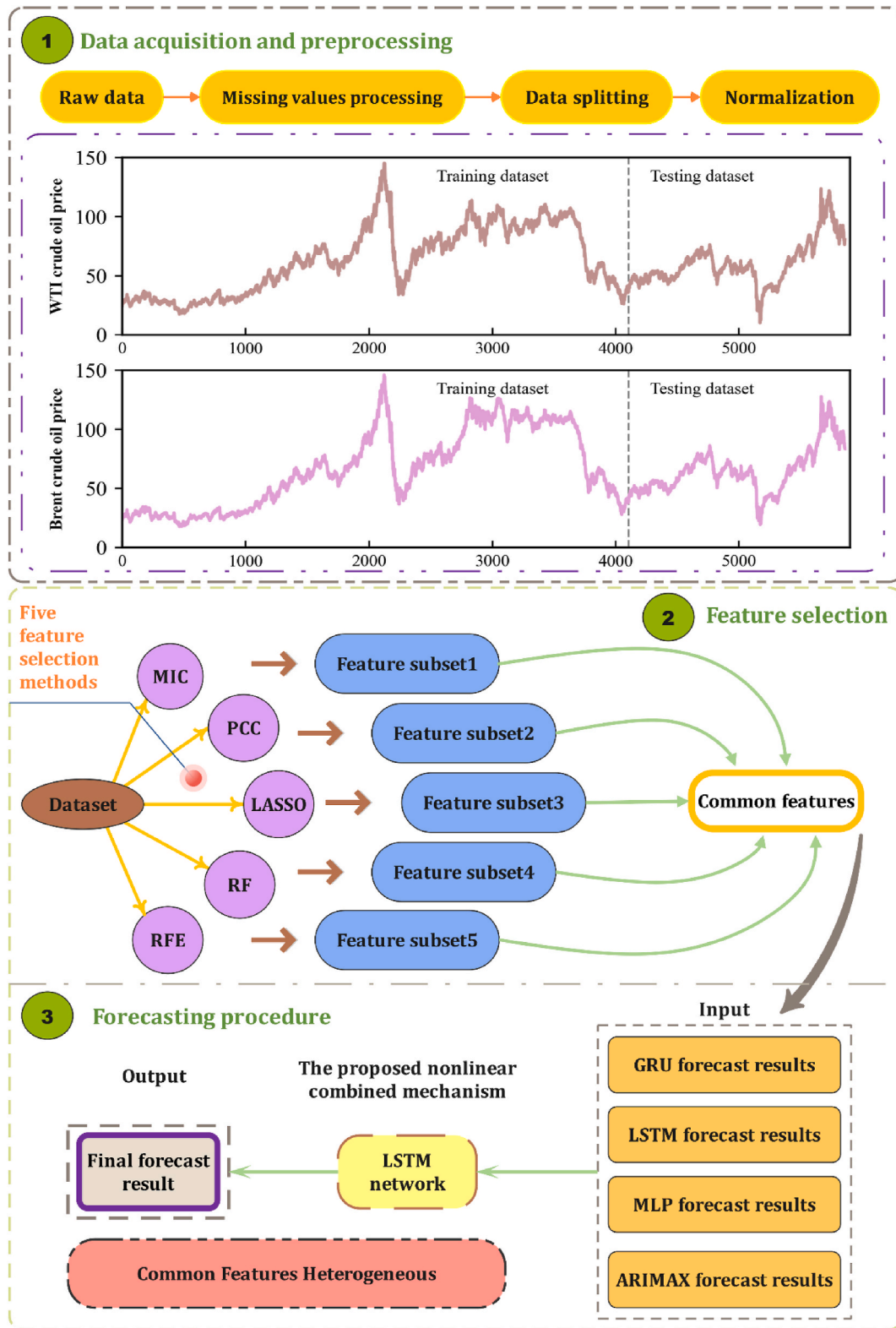


Fig. 1. The primary structure of the developed hybrid model.

oil market's supply and demand dynamics. Gold and silver, on the other hand, impact the crude oil market through various economic mechanisms, such as safe-haven demand, inflation expectations, and industrial use. These factors collectively contribute to a more nuanced understanding and prediction of crude oil price volatility.

The experiments use daily closing price series of WTI and Brent crude

oil as primary data. Exogenous variables include high-frequency data for five commodities (historical Brent prices, natural gas, historical WTI prices, gold, and silver), three major exchange rates (GBP/USD, USD/CNY, and EUR/USD), and two major stock market indices (S&P 500 and Dow Jones Industrial Average). In addition, the United States 10-year Treasury yield and the United States dollar index were used as proxies

for global macroeconomic volatility. As a result, the 11 high-frequency series used to forecast Brent and WTI crude oil prices were classified into four categories: commodities, foreign exchange, equities, and macroeconomic indicators. A detailed description of these characteristics is presented in Table 3.

The data spans from January 1, 2000, to November 30, 2022, with a total of 5865 data points from <https://www.investing.com/>. On April 20, 2020, the May 2020 contract for U.S. West Texas Intermediate (WTI) crude oil futures saw a historic -\$37.63/barrel close. In order to ensure as much forecast stability as possible, this outlier was corrected in this study to be populated with the April 21, 2020 closing price so that the minimum value of 10.01 instead of -\$37.63 appears in the statistical table.

To improve model performance, the data set is divided into two sections. The former 70% of the experimental data act as the training set and the remaining 30% constitutes the test set. Table 4 depicts the statistical message of datasets.

4.1.2. Feature selection

In this section, five feature selection methods: Pearson correlation coefficient (PCC), Maximal Information Coefficient (MIC), Random Forest Regressor (RF), LASSO and Recursive feature elimination (RFE), are applied during the experiment. Also, the WTI crude oil and Brent crude oil time series, as well as ten exogenous attributes, are introduced into the feature selection models mentioned above.

As illustrated in Table 5, it is evident that the variable P_{t-1}^{WTI} is contained in all feature subsets for the WTI crude oil time series, indicating the influence of oil price itself is endogenous for its prediction. A similar pattern is observed in the Brent crude oil time series. Additionally, there is a noteworthy relationship between these two oil prices, as each significantly affects the price of the other.

At the same time, silver, chosen by all selection mechanisms, also correlates closely with WTI crude oil and Brent crude oil. By the fact that silver and oil both have dollar-denominated assets, there exists a strong correlation within them. When oil price goes up, the increase of inflation follows closely, while silver prices also rise as inflation changes. Therefore, silver acts as a great attribute to multivariate analysis since it is a proper indicator of economic conditions. Also, the foreign exchange market plays a vital role in oil prices because exchange rates are the international metrics that illustrate the global economic situations. Thus, EUR-USD plays a vital role during the prediction process for WTI. Similarly, for Brent crude oil, USD-CNY is also considered an important factor, which is selected by all techniques.

In this study, in addition to focusing on the key indicators in the common variables, we find that the US Dollar Index (DXY) and Natural Gas (NG) are screened out in both datasets through a variety of methods.

Table 3

The description of various features.

Factor group	Name	Acronym
Commodities	WTI Crude Oil	WTI
	Brent Crude Oil	Brent
	Gold	GOLD
	Natural Gas	NG
	Silver	SV
Forex	GBP/USD exchange rate	GBP-USD
	EUR/USD exchange rate	EUR-USD
	USD/CNY exchange rate	USD-CNY
Stocks	S&P500 index	SPX
	Dow Jones Industrial Average Index	DJI
Macro	US 10 Year Bond Yield	US10B
	US Dollar Index	DXY

Note: There are both 10 exogenous attributes and its own past data for the WTI crude oil and Brent crude oil, which can be divided into four groups, namely Commodities, Forex, Stocks and Macro.

The undeniable point is that there is also some interpretability of features outside the set of common variables on crude oil prices.

4.2. Evaluation metrics

For the purpose of estimating the forecasting ability of models, some varying indicators are constructed.

- I. **RMSE:** It tends to measure the extent of differences between values of predicted and actual. In length, the indicators illustrate the standard deviation of realized values and forecasted ones. With regard to its being scale-dependent, RMSE is an excellent criterion for measuring accuracy in an effort to compare prediction errors of diverse models. Eq. (21) exhibits the equation.

$$RMSE = \sqrt{\frac{1}{T} \sum_{t=1}^T (P_t - \hat{P}_t)^2} \quad (21)$$

- II. **MAE:** MAE is another wonderful criterion for comparing predictions with final outcomes, allowing the average of the absolute errors to be accessed. The MAE equation is as follows:

$$MAE = \frac{1}{T} \sum_{t=1}^T |P_t - \hat{P}_t| \quad (22)$$

- III. **MAPE:** MAPE is introduced to ascertain the forecasting accuracy of a model. Described as a percentage, MAPE is constructed as shown in the following equation:

$$MAPE = \frac{1}{T} \sum_{t=1}^T |(P_t - \hat{P}_t) / P_t| \times 100\% \quad (23)$$

- IV. **R^2 :** Besides, the coefficient of determination, denoted as R^2 , is also employed to value models in these experiments.

$$R^2 = 1 - \frac{\sum_{t=1}^T (P_t - \hat{P}_t)^2}{\sum_{t=1}^T (P_t - \bar{P})^2} \quad (24)$$

In above formulas T is the size of sample, P_t and $\hat{P}_t$ are the actual and forecasted values at time t , respectively. Overall, while the values of these indexes are smaller, the prediction tends to be more precise.

Meanwhile, the parameters settings of the neural network model in all three experiments are given in Table 6.

4.3. Experimental setup

There are three experiments contained in this part, involving **Experiment I**, **Experiment II** and **Experiment III**. All experiments involved in this research are performed using Python 3.9 on a computing system running Windows 10. Python is utilized for data pre-processing, modeling, and prediction tasks. The computational environment also included Anaconda for package management and virtual environment handling, while development and debugging are conducted using PyCharm. The specific computing system used for the experiments is equipped an 11th Gen Intel® Core™ i5-11320H @3.20 GHz processor and 16.0 GB of RAM.

Specifically, the aim of **Experiment I** is to compare four single models, such as LSTM, ARIMAX, MLP, and GRU, with hybrid models. Specifically, all variables were incorporated into a single model for evaluation, yielding performance metrics for the single model.

Table 4

The statistical message of datasets.

Sample	Feature	Number	Max	Min	Mean	Std.
All Samples	WTI	5865	145.29	10.01	62.61	26.02
	Brent	5865	146.08	17.68	65.88	29.34
	GOLD	5865	2058.40	256.60	1054.80	519.14
	NG	5865	15.37	1.48	4.54	2.23
	SV	5865	48.59	4.03	16.01	8.41
	GBP-USD	5865	2.11	1.06	1.54	0.21
	EUR-USD	5865	1.59	0.82	1.19	0.15
	USD-CNY	5865	8.27	6.04	7.12	0.78
	SPX	5865	4796.56	676.53	1884.94	965.79
	DJI	5865	36799.65	6547.05	16383.96	7675.46
	US10B	5865	140.33	93.98	120.04	10.25
	DXY	5865	120.90	71.33	91.54	11.23
Training	WTI	4105	145.29	17.45	63.48	28.57
	Brent	4105	146.08	17.68	66.27	32.78
	GOLD	4105	1889.70	256.60	856.73	474.31
	NG	4105	15.37	1.63	5.03	2.29
	SV	4105	48.59	4.03	14.62	9.38
	GBP-USD	4105	2.11	1.37	1.65	0.17
	EUR-USD	4105	1.59	0.82	1.22	0.17
	USD-CNY	4105	8.27	6.04	77.29	0.87
	SPX	4105	2130.82	676.53	1344.98	331.76
	DJI	4105	18312.39	6547.05	11968.02	2710.24
	US10B	4105	135.66	93.98	116.89	9.85
	DXY	4105	120.90	71.33	89.41	12.48
Testing	WTI	1760	127.30	10.01	60.56	18.58
	Brent	1760	127.98	19.33	64.97	18.98
	GOLD	1760	2058.4	1129.8	1517.03	261.22
	NG	1760	9.64	1.48	3.38	1.57
	SV	1760	29.41	11.77	19.21	3.91
	GBP-USD	1760	1.48	1.06	1.31	0.06
	EUR-USD	1760	1.25	0.95	1.13	0.05
	USD-CNY	1760	7.31	6.26	6.71	0.24
	SPX	1760	4796.56	2000.54	3144.35	765.04
	DJI	1760	36799.65	17140.24	26683.65	5254.31
	US10B	1760	140.33	109.36	127.39	6.86
	DXY	1760	114.11	88.59	96.55	4.59

Table 5

Results of Features selection.

Dataset	Method	Feature Subset
WTI crude oil price	PCC	p_{t-1}^{WTI} p_{t-1}^{Brent} SV EUR-USD DXY USD-CNY GOLD US10B
	MIC	p_{t-1}^{WTI} p_{t-1}^{Brent} SV EUR-USD DXY USD-CNY
	RFE	p_{t-1}^{WTI} p_{t-1}^{Brent} SV EUR-USD DXY USD-CNY NG
	LASSO	p_{t-1}^{WTI} p_{t-1}^{Brent} SV EUR-USD NG SPX US10B
	RF	p_{t-1}^{WTI} p_{t-1}^{Brent} SV EUR-USD NG DXY US10B GBP-USD
	Common Feature	p_{t-1}^{WTI} p_{t-1}^{Brent} SV EUR-USD
Brent crude oil price	PCC	p_{t-1}^{WTI} p_{t-1}^{Brent} SV USD-CNY DXY EUR-USD GOLD
	MIC	p_{t-1}^{WTI} p_{t-1}^{Brent} SV USD-CNY DJI DXY
	RFE	p_{t-1}^{WTI} p_{t-1}^{Brent} SV USD-CNY NG GBP-USD EUR-USD
	LASSO	p_{t-1}^{WTI} p_{t-1}^{Brent} SV USD-CNY NG SPX EUR-USD
	RF	p_{t-1}^{WTI} p_{t-1}^{Brent} SV USD-CNY DXY SPX
	Common Feature	p_{t-1}^{WTI} p_{t-1}^{Brent} SV USD-CNY

Note: In this paper, there are two datasets employed. For WTI crude oil price forecasting, the Brent oil price is considered as the variable. Likewise, the prediction of WTI oil price also concludes the Brent oil price as the feature. Other nine common features are used by both datasets.

Subsequently, the prediction results of these single models are entered as variables into a nonlinear combination model, and LSTM, GRU, MLP, and CNN are used in **Experiment I** to complete the nonlinear combination. The best-performing model is finally selected for **Experiment III** to build a more accurate prediction model.

Table 6

Major parameters of prediction models.

Algorithm	Parameters	Value
CNN	Maximum iterations	80
	Hidden layers	2
	Batch size	32
	Validation split	0.05
GRU	Maximum iterations	100
	Hidden layer neurons	128
	Batch size	32
	Validation split	0.01
LSTM	Maximum iterations	100
	Dropout rate	0.05
	Batch size	32
	Validation split	0.05
MLP	Maximum iterations	50
	Dropout rate	0.01
	Batch size	32
	Validation split	0.01

Experiment II aims to ascertain whether the feature abstraction technique helps in building up forecasting accuracy or not. To avoid the contingency of a single approach, five different approaches are employed. Therefore, it introduces five selection methods, including MIC, PCC, RFE, LASSO and Random Forest. In this section, the prediction capability of the selected feature subsets is estimated by four models, i.e., LSTM, ARIMAX, MLP, and GRU. Additionally, the evaluation metrics for these models are compared with those obtained in **Experiment I**.

The purpose of **Experiment III** is to illustrate the effectiveness of the

hybrid model proposed and three combination methods are presented. In this section, this study aggregates the advantages of nonlinear combination approach and feature selection, which are exhibited in **Experiment I** and **Experiment II**, separately. As a consequence, three different combination mechanisms, namely, Single-subset heterogeneous combination, multi-subset homogenous combination and common-variable heterogeneous combination, are introduced.

4.4. Experiment I: comparison between single models and hybrid models

Table 7 describes the prediction results of four single models and four different hybrid models for the time series about WTI crude oil and Brent crude oil. In Fig. 2, the evaluation metrics value of MAPE and RMSE are illustrated. It can be observed that the hybrid models illustrate a much better performance when compared with the individual models, verifying the role of the novel nonlinear combination approach. Therefore, to compare the single models with the hybrid models, more details are indicated as follows.

- It is obvious that the four combined models outperform the single models on the whole, dramatically reducing the forecasting errors. For example, among single models for WTI, the ARIMAX captures the best forecasting accuracy with the lowest MAPE 4.9104%. However, three other hybrid models, namely CGRU, CMLP and CLSTM earn better forecasting results in comparison with the ARIMAX. Moreover, the other two evaluation metrics of MAE and RMSE also prove the validity of the hybrid models. It exactly verifies the ability of nonlinear hybrid models in identifying and grasping both the nonlinear and linear relationships. Moreover, the similar findings are reasonable to be concluded for Brent crude oil. Hence, it can be found that the nonlinear incorporation mechanism in this paper has great adaptability and stability.
- Especially, the hybrid model CLSTM earns the best overall forecasting accuracy with the superior values of evaluation metrics on two datasets. For WTI crude oil, in view of the indicator MAE, the biggest improvement rate reaches 47.69% in comparison with the individual model MLP. At the same time, compared with GRU, the reduction of MAE value is 2.0851. Furthermore, there is also a significant prediction ability improvement of CLSTM for Brent crude oil while comparing with other models. The above analysis shows that the nonlinear combination technique really plays a vital role in improving prediction precision. The reason for this phenomenon can be explained that the introduced combined method is able to avoid the occasionality of each single forecast. Therefore, it achieves the robustness of models for different metrics and datasets.
- In general, these four single models present good performance in the forecast of oil prices. However, incorporating the metrics of distinct individual models, four different nonlinear incorporation

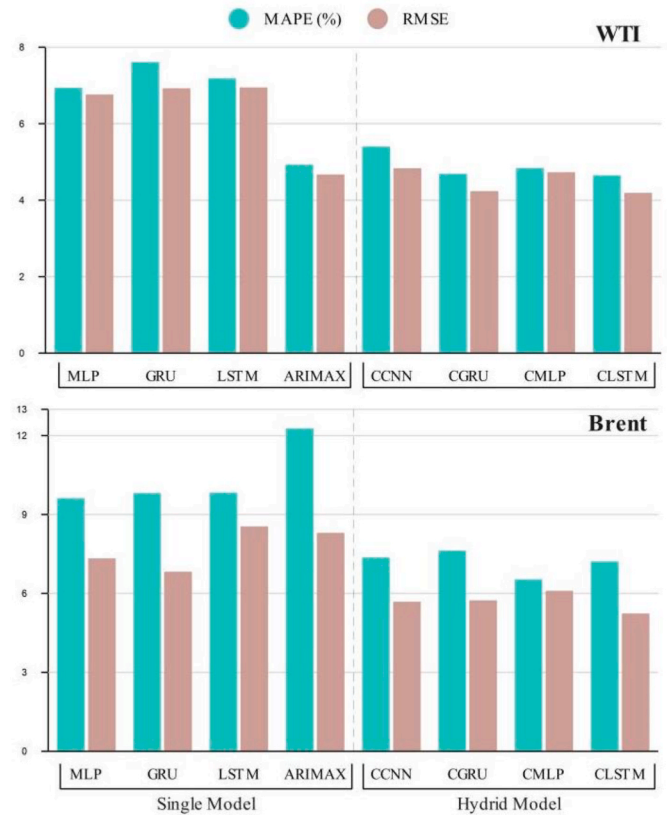


Fig. 2. Comparison between single models and hybrid models.

methods are developed, escaping the prediction occasionality and showing excellent forecast results on two datasets. Thus, it can be summarized that the recommended combination mechanism realizes the adaptability and stability of prediction. However, the performance of hybrid models may be potentially affected by their increased complexity. While they show better results in terms of accuracy, this complexity, if not properly managed, may lead to the risk of overfitting. Although the observed improvements are significant, their feasibility in practical applications needs further exploration.

4.5. Experiment II: comparison between models with and without feature selection technique

Experiment II is designated to test the effect of several feature selection methods, including PCC, MIC, RFE, LASSO, and RF, by comparing them with the models based on all features. There are also two datasets being chosen, WTI crude oil and Brent crude oil, as

Table 7
Comparison of performances of the single models and hybrid models.

Model		WTI crude oil price				Brent crude oil price			
		MAE	MAPE (%)	RMSE	R ²	MAE	MAPE (%)	RMSE	R ²
Single Model	MLP	4.8206	6.9199	6.7454	0.9108	5.8964	9.5979	7.3109	0.8515
	GRU	4.6067	7.5955	6.9061	0.9063	5.4791	9.7937	6.8047	0.8714
	LSTM	4.0459	7.1697	6.9345	0.9115	6.7647	9.8066	8.5271	0.7980
	ARIMAX	2.8713	4.9104	4.6516	0.9282	6.8146	12.2513	8.2763	0.8098
Hybrid Model	CCNN	3.2894	5.3891	4.8179	0.9335	4.3815	7.3413	5.6574	0.9110
	CGRU	2.6973	4.6735	4.2193	0.9491	4.4381	7.6047	5.7135	0.9093
	CMLP	3.0522	4.8231	4.7127	0.9363	4.6244	6.5047	6.0722	0.8975
	CLSTM	2.5216	4.6329	4.1746	0.9508	4.2164	7.1923	5.2087	0.9246

Note: Boldface refers to the best annual performance within its group in terms of MAE, MAPE, RMSE and R², respectively. The CCNN is combined by CNN, whose input are four previous single model forecast results. Similarly, the CGRU, CMLP and CLSTM using the same input as CCNN.

introduced. From Table 7, it is clearly observed that almost all models perform better while employing feature selection technology, significantly improving the evaluation metrics. The detailed prediction results are demonstrated in Table 8. Based on these results, the following conclusions can be drawn.

- (a) Firstly, in order to compare the models using feature selection technology with those based on all features, it is appropriate to take the MLP for example. For WTI crude oil, it can be found that all five methods using feature abstraction methods outperform the model based on all features. Especially through the analysis of evaluation metrics, it is concluded that the reduction of RMSE arrives at 56.38% for the MLP-PCC. This shows that the PCC technology helps in facilitating forecasting performance, the reason for which may be contributed to its effectiveness in capturing important information and reducing computation complexity. Furthermore, to avoid occasionality, five factor selection methods were introduced. However, the same conclusion is reasonable to be found, verifying the stability of the feature abstraction mechanism.
- (b) Then, to further ascertain the adaptability of the variable selection methods mentioned above, three other models GRU, LSTM, and ARIMAX are introduced in this subsection. Meanwhile, except for WTI crude oil, the Brent oil dataset is also applied to test its validity. Similarly, for these three combination models, the feature selection technologies also help in improving prediction effectiveness. For instance, the models with feature selections (i.e., GRU-RF and LSTM-RFE) yield better forecasting accuracy with MAPE 3.2116% and 4.4513%, respectively, whose reduction are 6.5821% and 5.3553% compared to the all-features data. Furthermore, in view of the Brent crude oil forecasting results, it is evident to draw a common conclusion. For example, compared with the model based on all-features data, MAPE was reduced by 3.1539%, 6.5821%, 5.3553% and 5.7422% for MLP-MIC, GRU-RF, LSTM-RFE and ARIMAX-RFE, separately. Hence, the above analysis further proves the validity and stability of feature selection methods in improving prediction precision.

- (c) In summary, the introduced feature selection method offers substantial advantages in enhancing prediction outcomes for the WTI and Brent datasets. However, it is essential to recognize that various models may exhibit sensitivities different from those of the selected feature subsets. Notably, the ARIMAX model demonstrates considerable performance improvements, but these gains may only be universally applicable across some models or tasks. Future research could explore additional model types.

4.6. Experiment III: comparison with the proposed model

In this section, the effectiveness comparisons of the single-subset heterogeneous method and multi-subset homogenous combination approach using LSTM are listed in Table 9 and Fig. 3. The single-subset heterogeneous approach tends to aggregate four prediction models for each variable abstraction technique. The multi-subsets homogenous combination method aims to assemble five feature selection approaches for each prediction method. A third approach, known as common feature heterogeneous, is introduced to develop a model (CM-CLSTM) with superior forecasting capabilities. What is worth mentioning is that it combines four forecasting models while using the features selected by all selection technologies. Obviously, the proposed model yields the best results within these two datasets discussed. More specific details are indicated below.

- (a) Among the single subset heterogeneous group, for WTI crude oil, the RFE-LSTM outperforms other models with MAE, MAPE, RMSE and R^2 1.7041, 3.1098, 2.8693%, and 0.9704, respectively. Since the single model can only recognize a fraction of components, this single-subset heterogeneous mechanism is proposed to offset the defect. Compared with the results illustrated in Experiments I and II, these four metrics are improved by a large rate. Meanwhile, the other four models, PCC-LSTM, MIC-LSTM, LASSO-LSTM and RF-LSTM, also realize better forecasting efficiency relative to the models in above two experiments. Thus, it can be found that the proposed incorporation mechanism aggregates the important information in distinct single models,

Table 8
Comparison of models based on different feature selection methods and all features.

Model	Method	WTI crude oil price				Brent crude oil price			
		MAE	MAPE (%)	RMSE	R^2	MAE	MAPE (%)	RMSE	R^2
MLP	PCC	1.8187	3.4199	2.9424	0.9708	4.3626	6.9878	5.3413	0.9207
	MIC	2.8547	3.4506	2.9001	0.9659	4.2570	6.4439	5.8291	0.9056
	RFE	2.3391	4.5021	3.3129	0.9672	4.5355	7.2329	6.0612	0.8979
	LASSO	3.9841	6.3925	5.7972	0.9142	5.7026	9.0959	7.2846	0.8526
	RF	2.9106	5.1046	3.9394	0.9555	5.9220	8.7164	7.0121	0.8634
	ALL	4.8206	6.9199	6.7454	0.9108	5.8964	9.5979	7.3109	0.8515
GRU	PCC	2.1118	3.4646	3.2734	0.9693	5.4447	8.6621	6.5084	0.8823
	MIC	2.9833	3.8314	3.0752	0.9629	5.5327	5.4207	5.0342	0.9296
	RFE	2.1787	4.3829	3.0822	0.9694	2.3507	3.8159	2.8904	0.9768
	LASSO	3.7674	6.9369	5.1031	0.9176	4.4220	6.2775	5.3575	0.9203
	RF	2.5665	4.3671	3.5682	0.9635	2.1119	3.2116	3.0372	0.9744
	ALL	4.6067	7.5955	6.9061	0.9063	5.4791	9.7937	6.8047	0.8714
LSTM	PCC	2.2796	4.5169	3.1666	0.9702	2.7797	4.6106	3.2619	0.9704
	MIC	2.5541	4.7171	3.9321	0.9557	3.7377	6.0505	4.2181	0.9506
	RFE	2.2321	4.1059	3.6505	0.9592	2.6281	4.4513	3.2462	0.9707
	LASSO	3.5687	6.6933	5.1258	0.9172	5.2979	7.3125	6.9356	0.8664
	RF	2.5229	4.4389	3.7362	0.9601	3.4225	5.8908	4.0166	0.9552
	ALL	4.0459	7.1697	6.9345	0.9115	6.7647	9.8066	8.5271	0.7980
ARIMAX	PCC	3.0688	5.1672	4.9132	0.9309	4.2434	7.0318	5.2415	0.9237
	MIC	3.1411	5.7601	5.0181	0.9279	6.7985	11.9825	8.3252	0.8075
	RFE	3.2492	6.0139	5.0041	0.9283	4.1683	6.5092	5.3898	0.9193
	LASSO	3.6017	6.1953	5.3074	0.9193	5.1866	9.1178	6.4235	0.8854
	RF	2.8837	4.9377	4.6455	0.9382	5.2753	9.2791	6.5900	0.8794
	ALL	2.8713	4.9104	4.6516	0.9282	6.8146	12.2513	8.2763	0.8098

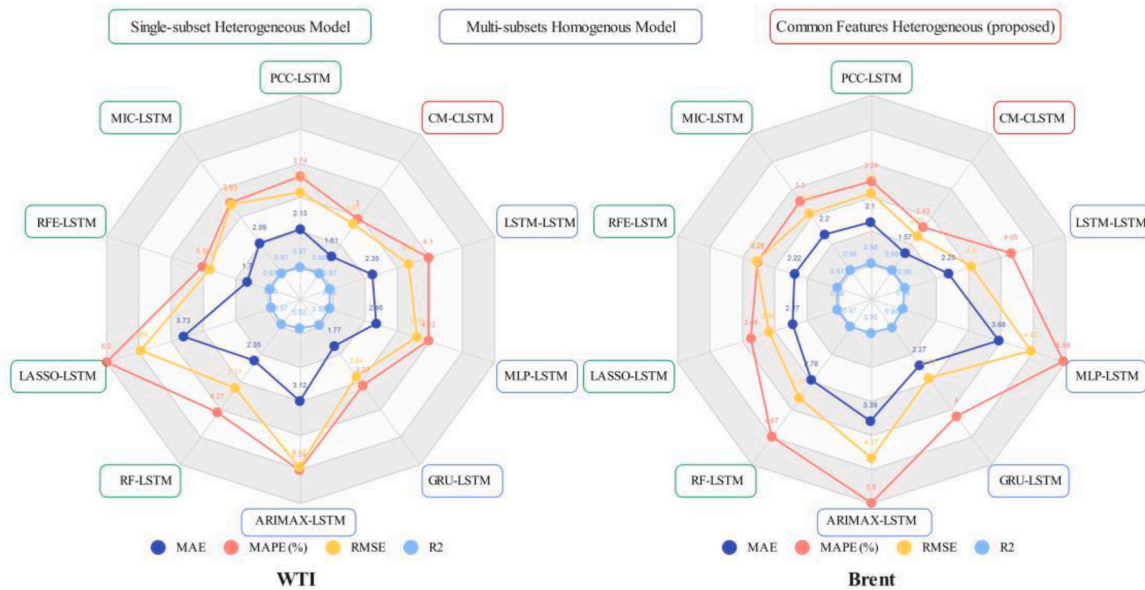
Note: Boldface refers to the best annual performance within its group in terms of MAE, MAPE, RMSE and R^2 , respectively.

Table 9

Comparison of single-subset heterogeneous combination approach, multi-subset homogenous approach and the proposed model.

Model	WTI crude oil price				Brent crude oil price			
	MAE	MAPE (%)	RMSE	R ²	MAE	MAPE (%)	RMSE	R ²
Single subset Heterogeneous								
PCC-LSTM	2.1297	3.7373	3.2551	0.9696	2.0959	3.2390	2.8987	0.9766
MIC-LSTM	2.0937	3.6125	3.5325	0.9671	2.1962	3.3042	2.9084	0.9765
RFE-LSTM	1.7041	3.1098	2.8693	0.9704	2.2190	3.2816	3.3021	0.9697
LASSO-LSTM	3.7344	6.2037	5.0921	0.9257	2.2711	3.4873	2.9562	0.9757
RF-LSTM	2.3495	4.2739	3.3664	0.9675	2.7829	4.6650	3.3688	0.9684
Multi subsets Homogenous								
ARIMAX-LSTM	3.1184	5.2386	5.1407	0.9243	3.3874	5.6023	4.3654	0.9470
GRU-LSTM	1.7657	3.2676	2.9368	0.9753	2.2722	4.0031	2.7282	0.9793
MLP-LSTM	2.4554	4.1207	3.7553	0.9596	3.6755	5.5561	4.6249	0.9405
LSTM-LSTM	2.3520	4.1011	3.4638	0.9657	2.2472	4.0549	2.8954	0.9767
Common Features Heterogeneous (proposed)								
CM-CLSTM	1.6135	2.9962	2.8079	0.9774	1.5651	2.4314	2.1459	0.9797

Note: Boldface refers to the best performance within its group in terms of MAE, MAPE, RMSE and R². All these models are combined by LSTM network. The single-subset heterogeneous approach combines four forecast results produced by ARIMAX, GRU, MLP and LSTM under each feature selection method. For ARIMAX-LSTM, the multi-subsets homogenous combines five forecasting results based on ARIMAX, using five feature selection technologies. The proposed CM-CLSTM incorporates four single model forecasting results, including ARIMAX, GRU, MLP and LSTM, which is based on common features obtained by five feature selection mechanisms.

**Fig. 3.** Comparison of two combination methods with the developed hybridization.

successfully. Furthermore, in terms of the Brent crude oil, it is easy to draw the same conclusion, showing the robustness and stability of this combination technique.

- (b) Among the multi-subset homogenous group, the GRU-LSTM acquires a superior forecasting performance for WTI crude oil with the lowest value of MAPE 3.2676%. Meanwhile, three other evaluation indicators of MAE, RMSE and R² also present the validity of the hybrid model mentioned above. Given that the single variable selection method can only recognize a part of information, this multi subsets heterogeneous mechanism is proposed to escape this deficiency. In comparison with the performance of models in **Experiments I and II**, there is a significant improvement in prediction effectiveness. Therefore, the new combination method succeeds in incorporating different elements of five feature selection technologies. In addition, in view of the Brent crude oil dataset, the similar found can be summarized, thus verifying the adaptability and validity of the multi subsets heterogeneous mechanism.

- (c) Finally, aiming to improve the stability and effectiveness of forecasting further, a novel common features heterogeneous incorporation model, namely CM-LSTM, is developed. The MAPE of the proposed model is the lowest compared with RFE-LSTM and GRU-LSTM, along with the other two error metrics. The possible reason is that this new method aggregates the advantage of feature selection, and the combined model, so it presents a great performance in terms of precise and robustness. However, despite the excellent performance of CM-LSTM on the current dataset, one still needs to be wary of the stability of its performance under different market fluctuations and external disturbances. In addition, the increased complexity of the model may lead to a rise in training time and resource consumption, which also needs to be considered in practical applications. Therefore, future research should further optimize the model structure to balance accuracy and computational efficiency.

4.7. Summary: based on experiments I–III

Based on the analyses of **Experiment I** (Comparison between single models and hybrid models), **Experiment II** (Comparison between models with and without feature selection technique) and **Experiment III** (Comparison with the developed model), this work can draw the following conclusions.

- Nonlinear combination models perform better in terms of prediction accuracy than single models. Among them, the nonlinear combination approach based on LSTM models performs particularly well.
- The prediction after feature screening outperformed the prediction using all features. This trend was demonstrated in various models on both datasets, including MLP, CNN, LSTM, and ARIMAX. The data analyses suggest that feature filtering on the current dataset helps to reduce redundant information, which improves the prediction accuracy and efficiency of the models.
- The proposed nonlinear combinatorial model based on common variables as well as LSTM not only has stronger prediction ability compared with single-subset homogeneous model and multi-subset heterogeneous model but also significantly improves the prediction accuracy.

Finally, compared to the benchmarks, the developed model exhibits superior performance according to four evaluation criteria. This improvement suggests that significant resources and economic costs could be saved if the Mean Absolute Percentage Error (MAPE) is reduced. Based on the previous description and conclusions in this subsection, the proposed hybrid model is convincingly robust, highly accurate, and practical for forecasting crude oil prices.

4.8. Trading simulations

After developing reliable forecasting models, it is important to consider how forecasts can be effectively used to inform practical decisions and optimize trading and investment activities. A key aspect when trading and investing is the estimation of expected returns. Based on the estimated returns, an investment strategy can be developed. The profitability is calculated as follows:

$$\hat{R}_t = \frac{\hat{P}_t - \hat{P}_{t-1}}{P_t} \quad (25)$$

where $\hat{P}_t$ and $\hat{P}_{t-1}$ are the forecasted values at time t and $t-1$, respectively. And the $\hat{R}_t$ is the predicted return rates of forecasts at time t .

The specific trading strategy setup for the simulated quantitative investment is described below, where daily returns are assumed to remain constant and then annual returns can be calculated.

Scenario: The strategy trades based on the prediction of future returns. The core idea is to use a deep learning model to predict the future returns of crude oil prices to realize positive returns and avoid negative returns. 50% of the current capital will be used to buy when the predicted future closing price is higher than the current closing price; conversely, all positions will be sold when the predicted future closing price is lower than the current closing price. Each trade will incur a transaction fee of 0.1%. In addition, the risk-free rate of return is set at 5%.

Table 10 shows the simulated trading results for different models. It can be seen that the CM-CLSTM model-based trades perform prominently. In the WTI dataset, the CM-CLSTM model has an annualized return of 22.45% and a Sharpe ratio of 0.5060 in the simulated investment trades, which is better than the other benchmark models. In contrast, the ARIMAX-LSTM model performs best in the Brent dataset, with an annualized return of 61.57% and a Sharpe ratio of 1.8754. The CM-CLSTM model, although ranked fourth in the Brent dataset, also shows some advantages with an annualized return of 27.29% and a Sharpe ratio of 0.9289.

The proposed CM-CLSTM model shows a relatively good trading performance overall. However, the adaptability of the model on different datasets needs to be further improved. In particular, the performance in specific market environments, such as the Brent dataset, still has a gap compared to the ARIMAX-LSTM model. Further optimization of the model is needed in future research to enhance its stability and effectiveness under various market conditions.

5. Discussion

An insightful discussion is contained in the following, so as to further ascertain the superiority of proposed model.

5.1. Rolling forecasting

In order to verify the robustness of the proposed prediction model and further evaluate the generalization performance of the model, this study adopts a rolling prediction process, where the dataset is divided into ten data subsets according to chronological order. As shown in **Fig. 4**, the dataset contains a total of 586 days, with three rolling test periods: the first period is from April 4, 2016, to June 24, 2018, the second period is from June 25, 2018 to September 9, 2020, and the third period is from September 10, 2020 to November 30, 2022.

Three rounds of experiments are conducted, and the final results in this subsection and the Experiment III in **subsection 4.6** are presented in **Table 11**. According to the results in **Tables 11** and it can be seen that the proposed model continues to show strong prediction performance, significantly outperforming the benchmark models in **Experiments I, II, and III**, exhibiting the same level of prediction as the proposed model in Experiment III, further demonstrating that the proposed prediction

Table 10
The simulated trading results.

WTI crude oil price				Brent crude oil price			
Model	Annualized Return (%)	Annualized Volatility (%)	Sharpe Ratio	Model	Annualized Return (%)	Annualized Volatility (%)	Sharpe Ratio
CM-CLSTM	22.4504	43.3811	0.5060	ARIMAX-LSTM	61.5696	32.5635	1.8754
MIC-LSTM	19.6973	42.7179	0.4494	LASSO-LSTM	34.5938	25.0146	1.3630
RFE-LSTM	19.6882	43.0509	0.4457	RF-LSTM	7.9232	6.8515	1.0834
LSTM-LSTM	17.4603	42.3688	0.4003	CM-CLSTM	27.2924	28.8443	0.9289
LASSO-LSTM	18.7469	48.0896	0.3794	PCC-LSTM	26.0474	29.9887	0.8519
GRU-LSTM	15.9210	44.9706	0.3429	MIC-LSTM	15.1848	18.1492	0.8091
ARIMAX-LSTM	12.3233	42.0955	0.2809	LSTM-LSTM	27.8491	34.1480	0.8009
PCC-LSTM	7.2412	39.1660	0.1721	MLP-LSTM	8.3507	11.1933	0.7014
MLP-LSTM	2.1327	36.6634	0.0445	GRU-LSTM	24.8962	39.2532	0.6215
RF-LSTM	1.9010	32.8107	0.0427	RFE-LSTM	−1.0152	25.7690	−0.0588

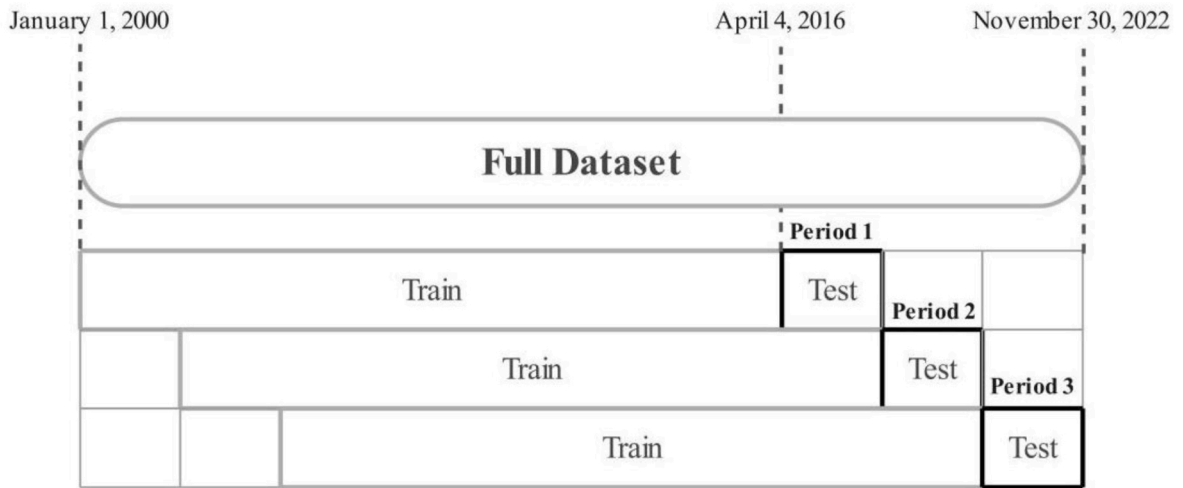


Fig. 4. Rolling forecasting process.

Table 11

The rolling forecasting results of the proposed CM-CLSTM model.

Period	WTI crude oil price				Brent crude oil price			
	MAE	MAPE (%)	RMSE	R ²	MAE	MAPE (%)	RMSE	R ²
1	1.0228	1.9963	1.3016	0.9714	1.0738	1.9721	1.3753	0.9781
2	1.6832	4.3318	3.4400	0.9356	1.4338	2.7249	1.9318	0.9817
3	2.2316	2.8245	3.1693	0.9772	2.2411	2.6387	3.2072	0.9781
Average	1.6459	3.0509	2.6370	0.9614	1.5869	2.4572	2.1721	0.9793
Experiment III	1.6135	2.9962	2.8079	0.9774	1.5651	2.4314	2.1459	0.9797

model can exhibit strong and stable prediction accuracies, even across different datasets.

5.2. Diebold–Mariano (DM) test

To evaluate the effectiveness of the proposed model aforementioned, the Diebold–Mariano test, a widely employed test approach, is introduced appropriately. The main steps are presented as follows:

$$H_0 : E[P(r_t^1)] = E[P(r_t^2)] \quad (26)$$

$$H_1 : E[P(r_t^1)] \neq E[P(r_t^2)] \quad (25)$$

The above null hypothesis assumes that there is not much difference between the prediction effectiveness of these two models, while the alternate hypothesis supposes that the difference is remarkable. The loss function P is the forecasting errors function. And the r_t^1 and r_t^2 are the predicted errors between actual values and forecasts of distinct models.

$$\bar{e} = \frac{1}{D} \sum_{t=1}^D [P(r_t^1) - P(r_t^2)] \quad (27)$$

In the above equation (27), D denotes the length of forecasting values. $\bar{e}$ is the mean of the loss function difference of two models.

$$DM = \frac{\bar{e}}{\sqrt{2\pi f_e(0)/D}} \rightarrow N(0, 1) \quad (28)$$

The DM test theory assumes that DM statistic conforms to the normal distribution and the $f_e(0)$ presents the zero spectral density. When the significance level is α , the rejection domain is $W = \{|DM| > |z_{\alpha/2}|\}$. The original hypothesis is to be rejected if the DM statistic falls into the rejection domain. And there is prominent difference in the predictive power of the two models. For instance, if $|DM| > 1.96$, at a 5% level the

original hypothesis is rejected.

In Table 12, the test results of comparison between the proposed models and other models are depicted. For the result in Experiment I, the

Table 12

The Diebold–Mariano test statistic values (P values) of diverse models.

Model	WTI crude oil price	Model	Brent crude oil price
Experiment I			
MLP	16.0901 (0.0000)	MLP	3.8622 (0.0001)
GRU	15.9296 (0.0000)	GRU	1.4521 (0.1466)
LSTM	9.7043 (0.0000)	LSTM	4.3855 (0.0000)
ARIMAX	8.8773 (0.0000)	ARIMAX	32.6586 (0.0000)
CCNN	18.4335 (0.0000)	CCNN	18.5793 (0.0000)
CGRU	23.4245 (0.0000)	CGRU	20.9290 (0.0000)
CMLP	20.6547 (0.0000)	CMLP	19.6317 (0.0000)
CLSTM	15.2241 (0.0000)	CLSTM	21.0777 (0.0000)
Experiment II			
MLP-PCC	19.0452 (0.0000)	MLP-PCC	6.3850 (0.0000)
GRU-PCC	9.6169 (0.0000)	GRU-RFE	8.8973 (0.0000)
LSTM-RFE	3.2713 (0.0010)	LSTM-RFE	9.2937 (0.0000)
ARIMAX-RF	17.3371 (0.0000)	ARIMAX-RFE	20.7061 (0.0000)
Experiment III			
PCC-LSTM	13.2332 (0.0000)	PCC-LSTM	13.8020 (0.0000)
MIC-LSTM	8.0687 (0.0000)	MIC-LSTM	16.4099 (0.0000)
RFE-LSTM	2.5193 (0.0118)	RFE-LSTM	6.8973 (0.0000)
LASSO-LSTM	34.1150 (0.0000)	LASSO-LSTM	16.4714 (0.0000)
RF-LSTM	14.7552 (0.0000)	RF-LSTM	8.3347 (0.0000)
ARIMAX-LSTM	19.8648 (0.0000)	ARIMAX-LSTM	25.2688 (0.0000)
GRU-LSTM	5.0003 (0.0000)	GRU-LSTM	11.3517 (0.0000)
MLP-LSTM	18.0875 (0.0000)	MLP-LSTM	23.0030 (0.0000)
LSTM-LSTM	17.5561 (0.0000)	LSTM-LSTM	4.2206 (0.0000)

Note: The table presents the results of the Diebold–Mariano test, containing z-statistic values and P-values. The compared models involve nearly all models in three experiments.

P values provided are all much lower than 0.05, showing that the feature selection approaches play a vital role in improving forecasting ability. Looking into the *P* values in **Experiments II** and **III**, it is evident that the proposed model also presents a better performance. Therefore, it is distinct that the novel proposed combination method surpasses other compared models. In brief, this subsection validates that the proposed hybrid model captures superior forecasting accuracy and robustness, once again.

5.3. Improvements of the hybridization

In this subsection, an improvement percentage of the MAPE criteria, aiming to exhibit a convictive analysis of the proposed model, is adopted. The equation is defined as:

$$I_{MAPE} = \frac{E_a - E_b}{E_a} \times 100\% \quad (29)$$

In Eq. (29), E_b is the MAPE value of the developed model and E_a is the MAPE value of the compared model. The value of I_{MAPE} denotes the extent of improvement by the proposed model. If the value is lower than zero, then the compared model performs better.

To further evaluate the forecasting advantage of the hybridization, whose prediction results are compared with models introduced in these three experiments, the improvements of MAPE are depicted within **Table 13**. It is meaningful to exhibit some strict and detailed analysis, as presented below.

- (a) In comparison with the models involved in **Experiment I**, the largest improvement percentage achieved at 60.55% for WTI and that's of the Brent time series is 60.32%. Meanwhile, the minimum percentage is also as high as 35.33% in the field of WTI crude oil. Furthermore, it is visible that the performance of four combined models exceeds those of single methods on the whole, regardless of the dependent and the model used. Consequently, there is no denying that the effect of feature selection and nonlinear combination approach really makes sense.
- (b) Referring to those models do well relatively within **Experiment II**, the developed hybridization realizes improvement as usual. It leads to reductions of 39.32% and 27.03% compared with

ARIMAX-RF and LSTM-RFE for WTI crude oil, respectively. Similarly, in the Brent time series, the maximum improvement percentage reached at 48.04%, which is not easy to ignore. Nevertheless, there is also an exception, which is a common thing. GRU-RFE and LSTM-GRU has a slightly lower value of MAPE than the developed model. However, other evaluation indicators, including MAE, RMSE and R^2 , of the proposed hybridization also perform better. Thus, it can further be recognized that the nonlinear combination approach helps facilitate prediction preciseness.

- (c) With respect to the models in **Experiment III**, which introduce different combination methods after feature selection, it is marked to us that they outperform those of the two experiments mentioned above. However, the proposed model still achieved better effectiveness, with the biggest improvement percentage of 42.81% and 48.60% for the WTI and Brent time series, respectively. At the same time, the minimum improvement also attains 3.651% about WTI crude oil, while that's of the Brent is 3.201%. In sum, the developed hybridization succeeded in obtaining satisfactory forecasting accuracy.

5.4. Explanation of the output via SHAP

5.4.1. Model contribution analysis

This section employs Shapley additive interpretation (SHAP) to analyze complex crude oil price forecasting models to increase confidence in the forecast results. These techniques assist decision-makers by clarifying the influence and contribution of each input variable on the final output. Brent crude oil is used as an example to analyze forecast results utilizing SHAP.

Based on the framework of this study, this section will be analyzed in two sub-modules: model importance and variable importance. Firstly, the interpretability method of SHAP is applied to the final LSTM model to assess the contribution of the four models used for the nonlinear combination. **Fig. 5** demonstrates the results of the analyses, and this study calculates the contribution based on the average absolute SHAP values. As shown in **Fig. 5(a)**, the ARIMAX model has the highest contribution, followed by the LSTM, with both having almost equal contributions, followed by GRU and MLP. **Fig. 5(b)** illustrates the Shapley values for each model in each sample, which allows for observing the extent to which these models have an impact on the dataset. The y-axis of the graph represents the models, the x-axis represents the SHAP values, and the dots are coloured from blue to red to indicate the change in model forecasting values. It can be observed that the higher the predicted value of the model, the higher the overall SHAP value, with a consistent trend across the four models.

5.4.2. Feature importance analysis

In the framework of this study, the treatment of variables is centred around two main aspects: feature screening, discussed in **subsection 4.1.2**, and model prediction, covered in **subsections 4.4 to 4.6**. During feature screening, five distinct methods are employed, identifying five unique feature subsets. This subsection determines the core dataset by analyzing the intersection of these subsets, i.e., the common variables. The final results (shown in **Table 5**) indicate that commodity factors and exchange rates are crucial core features for both WTI and Brent crude oil price data. This suggests a strong correlation between WTI and Brent crude oil prices, as historical data for both are included in the subset of common variables. Additionally, silver prices emerged as the second core influence, likely due to a market correlation between silver and crude oil prices, which may be linked to their shared role in the global economy and investors' risk-averse behaviour towards these assets.

Nevertheless, the common variables for WTI and Brent crude oil differ in terms of exchange rates. For WTI, the EUR-USD exchange rate is included as the final variable in the common subset. This could be attributed to the characteristics of the WTI market and its relationship

Table 13
Improvement percentage of the developed model.

Experiments	Model	WTI	Model	Brent
Experiment I	MLP	54.05%	MLP	74.43%
	GRU	60.55%	GRU	74.94%
	LSTM	58.21%	LSTM	74.97%
	ARIMAX	38.98%	ARIMAX	79.97%
	CCNN	44.40%	CCNN	66.57%
	CGRU	35.89%	CGRU	67.73%
	CMLP	37.88%	CMLP	62.27%
Experiment II	CLSTM	35.33%	CLSTM	65.88%
	MLP-PCC	12.39%	MLP-PCC	64.88%
	GRU-PCC	13.52%	GRU-RFE	35.69%
	LSTM-RFE	27.03%	LSTM-RFE	44.87%
	ARIMAX-RF	39.32%	ARIMAX-RFE	62.30%
Experiment III	PCC-LSTM	19.83%	PCC-LSTM	24.23%
	MIC-LSTM	17.06%	MIC-LSTM	25.72%
	RFE-LSTM	3.651%	RFE-LSTM	25.21%
	LASSO-LSTM	51.70%	LASSO-LSTM	29.63%
	RF-LSTM	29.90%	RF-LSTM	47.39%
	ARIMAX-LSTM	42.81%	ARIMAX-LSTM	56.19%
	GRU-LSTM	8.311%	GRU-LSTM	38.69%
	MLP-LSTM	27.29%	MLP-LSTM	55.83%
	LSTM-LSTM	26.94%	LSTM-LSTM	39.48%

Note: This table illustrates the improvement percentage of the developed model compared with other models. There are two values are negative, denoting the compared model performs better, which is a common situation.

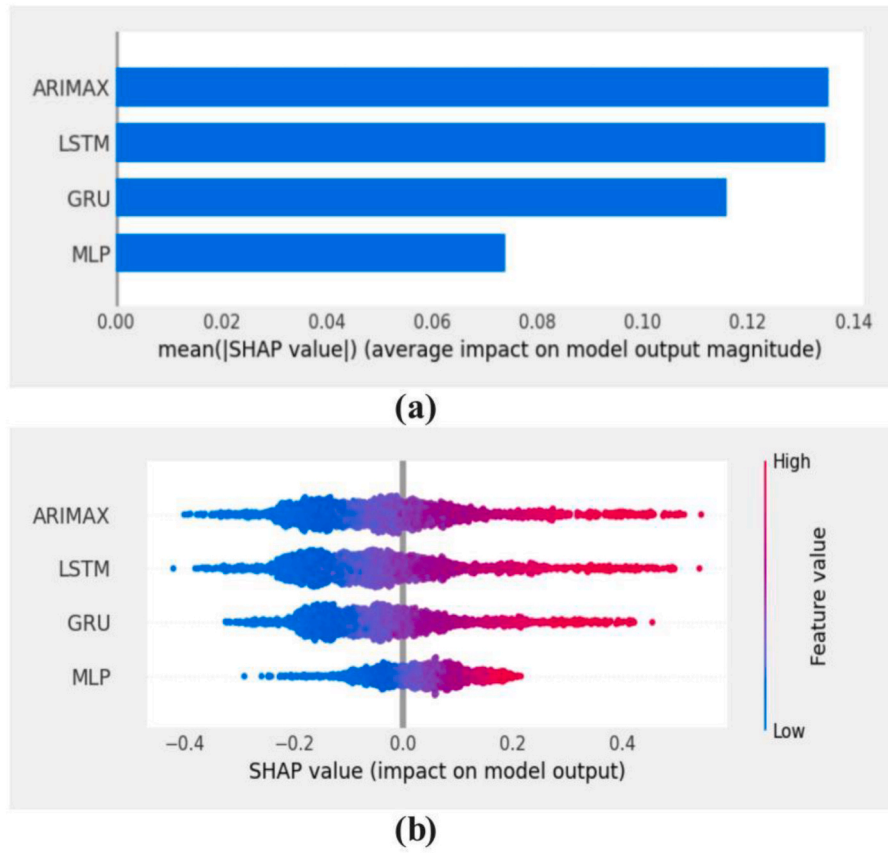


Fig. 5. SHAP summary plot:(a) The global importance of the input model; (b) The SHAP value of each of model for each sample.

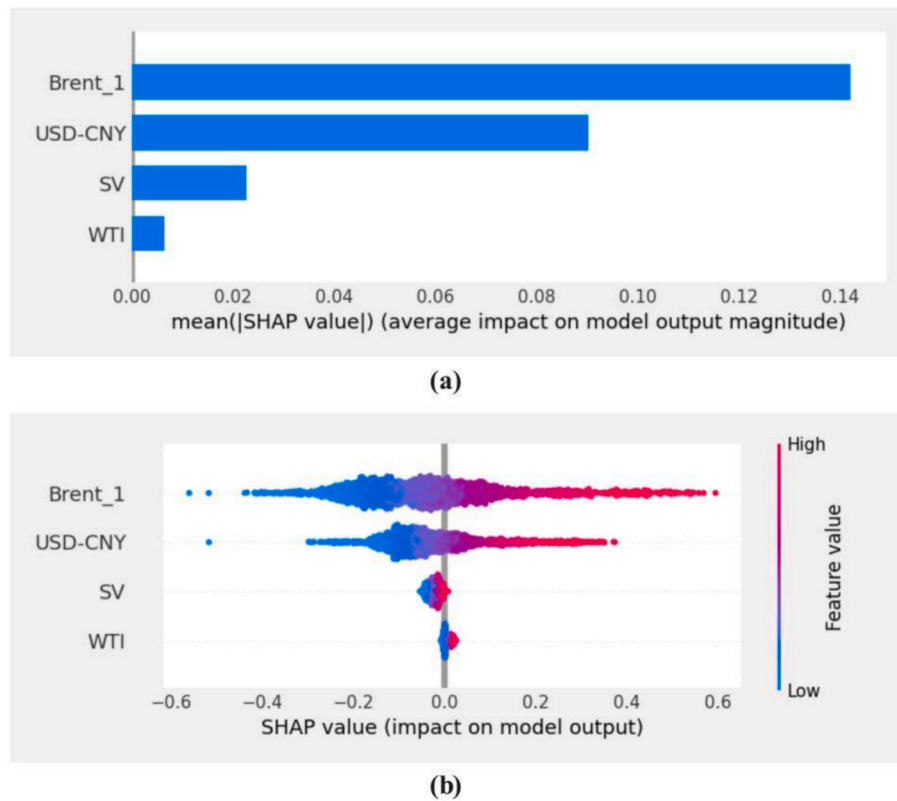


Fig. 6. SHAP summary plot for LSTM model:(a) The global importance of the input feature; (b) The SHAP value of each of feature for each sample.

with the USD exchange rate. Since WTI crude oil is predominantly priced in US dollars, fluctuations in the USD exchange rate significantly impact WTI prices, and changes in the EUR-USD exchange rate may indirectly reflect trends in the USD exchange rate. Conversely, the USD-CNY exchange rate is identified as a core influence for Brent crude oil. This may be because changes in the RMB exchange rate affect the demand for crude oil in China, a major consumer, which in turn impacts the price of Brent crude oil. Therefore, fluctuations in the USD-CNY exchange rate significantly affect the volatility of Brent crude oil prices.

This study employs a SHAP approach in the model prediction section to quantitatively analyze the overall contribution of common features in the prediction process. While the contribution of the model in the final non-linear combination approach has been previously analyzed, this study advances by examining the contribution levels of variables in the prediction process using an LSTM as an example. Fig. 6(a) illustrates the overall contribution levels of the variables, while Fig. 6(b) demonstrates how the features influence the dataset.

Fig. 6(a) shows that the closing price of historical trading data has the highest impact on forecasting the future closing price of Brent. Secondly, USD-CNY also has a considerable impact on the exchange rate prediction results, followed by silver, and WTI historical data has the most negligible impact in relative terms; Fig. 6(b) plots the Shapley values for each feature for each sample, from which it is possible to see which features are the most important and how much they affect the dataset. The y-axis represents the features, while the x-axis shows the SHAP values corresponding to each feature. The colour of the dots ranges from blue, indicating smaller feature values, to red, indicating larger feature values. As illustrated in Fig. 6(b), higher values of Brent historical closing prices, USD-CNY exchange rates, and silver prices are associated with an increased forecast value. Conversely, the impact of historical WTI data on Brent's forecast is inversely related.

6. Conclusion

Crude oil price forecasting has garnered significant attention and research interest due to its substantial impact on various economic sectors. Accurate forecasting aids in making informed policy decisions and economic planning. However, the fluctuation of oil prices is influenced by a range of factors, making price prediction a challenging endeavor. Traditionally, oil market fundamentals are widely used in forecasting. However, given the limitations of real-time access to fundamental information, utilizing high-frequency financial data has become increasingly valuable. These financial data are less likely to be revised and provide more detailed information. Consequently, it introduces eleven financial features for WTI and Brent crude oil price prediction. This research presents a novel hybrid model that combines feature extraction techniques with a nonlinear combination mechanism. The experimental results demonstrate that this new model achieves superior accuracy and robustness.

Through the discussion above, more conclusions are drawn as follows: (1) It is evident to verify the ability of different feature selection technologies within information extraction. Furthermore, the common message is of great significance in improving the effectiveness of prediction under the role of different selection methods. (2) The forecasting effectiveness of the nonlinear hybrid mechanism, which capitalises nonlinear information of different single models, definitely outperforms individual model. It can be found that it avoids the occasionality of one single model and owns the robustness. (3) the hybridization proposed, which successfully associates the merits of feature extraction and nonlinear combination approach, earns superior accuracy and stability compared with the other two combination methods.

Future research can be improved in several ways to enhance the effectiveness of the model. The current model still needs to be improved in terms of domain-specific data adaptability, anomalous data handling capabilities, and model interpretability. Future research should prioritize addressing these issues to enhance and optimize model

performance. Second, the predictive ability of the model can be enhanced in the future by introducing more relevant data sources, such as combining macroeconomic indicators with sentiment and public opinion data to provide more comprehensive contextual information in order to improve the accuracy and robustness of the model. Further, the current analyses need to consider the time-varying effects of variables more fully. Future research will aim to conduct more detailed time-varying analyses to capture better and understand the dynamic effects of variables over time.

Overall, the proposed hybrid model offers a new and reliable method for forecasting oil prices in real-time using only financial data. It demonstrates significant predictive accuracy and effectiveness and provides notable convenience and timeliness, which is valuable for both practitioners and researchers. Future research could explore applying this model to forecast other asset prices. Additionally, investigating more advanced integration methods could be beneficial.

CRediT authorship contribution statement

Yan Xu: Funding acquisition, Investigation, Methodology, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. **Tianli Liu:** Data curation, Investigation, Software, Validation, Writing – review & editing, Writing – original draft. **Qi Fang:** Data curation, Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **Pei Du:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Supervision, Validation, Writing – original draft, Writing – review & editing. **Jianzhou Wang:** Conceptualization, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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